

# A Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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## Abstract

We determine the differential-module and resonance structure of axial Schwarzschild perturbations in four-dimensional pure Weyl gravity. The six-state Bach system has an exact filtration with successive quotients given by two spin-two Regge–Wheeler modules and one spin-one Maxwell module. The repeated spin-two block is a non-split self-extension and is the first jet of a four-state spin-two/spin-one family.

After Liouville reduction, the extension is represented by a scalar projective cocycle modulo the symmetric-square operator

$$\mathcal{K}_U = D^3 + 4UD + 2D(U).$$

For axial  $\ell = 2$  we prove the exact normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[ 1 - \frac{2}{r} \right].$$

The class in brackets is exactly the transverse-traceless critical spin-two mass jet. Thus the local parameter relation is  $m = (i\omega/2)\tau$ , modulo rational triangular gauge. The required gauge grows linearly at infinity; an explicit boundary transgression therefore remains before the globally normalized Jost slopes can be identified.

At a simple spin-two quasinormal frequency the complete connection admits two possible local Smith types. A validated projective Evans calculation selects the defective type  $(0, 0, 2)$  at one damped Schwarzschild mode. The geometric root vector is Einstein, whereas its generalized root vector has nonzero projection to the Ricci-carrier quotient. We further show that the compact-source, compact-observation outgoing Green operator on the Schwarzschild exterior has a nonzero rank-one pole of order two at this frequency.

Promotion to a global physical resolvent and to a time-domain generalized ringdown term requires a separate analytic Fredholm realization, weighted endpoint domains, and bounded metric reconstruction.

## 1 Introduction and main results

Pure Weyl gravity has a fourth-order metric equation, but its Schwarzschild perturbations are not best understood as an unstructured fourth-order system. In the axial  $\ell = 2$  sector the six-dimensional radial system has the filtered form

$$0 \subset M_2 \subset E_2 \subset M_{\text{Bach}}, \quad \text{gr } M_{\text{Bach}} \simeq M_2 \oplus M_2 \oplus M_1,$$

where  $M_2$  is the spin-two Regge–Wheeler module and  $M_1$  is the spin-one Maxwell module. The middle object  $E_2$  is a self-extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0.$$

The decisive point is that this extension is non-split over the rational differential field. The two spin-two layers therefore share scalar characteristics without defining two rationally separable radial branches.

This paper establishes four central results.

1. The repeated spin-two block is a non-split Regge–Wheeler self-extension, realized exactly as a parameter jet.
2. Its projective extension class has the elementary normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[ 1 - \frac{2}{r} \right].$$

3. The class  $[1 - 2/r]$  is exactly the transverse-traceless critical mass jet. The local equivalence gauge has forced linear growth at infinity, and its failure to preserve Jost frames is an explicit boundary Wronskian.
4. At a certified damped Schwarzschild quasinormal frequency the connection is defective. The associated generalized root vector has a nonzero carrier quotient, and the exterior cut-off outgoing Green operator has a genuine rank-one double pole.

The third result is stronger than a statement about a finite connection matrix. It is a meromorphic differential-response theorem for compact radial sources and local radial observations. It is deliberately weaker than a causal spacetime resolvent theorem: no inverse-Laplace deformation, late-time expansion, or  $te^{i\omega_n t}$  assertion is made here.

## Logical status

| Result                                         | Required input                                                  | Status                      |
|------------------------------------------------|-----------------------------------------------------------------|-----------------------------|
| RW/RW/Maxwell filtration and partial jet       | Exact rational identities and reconstruction rank               | Exact                       |
| Non-split spin-two extension                   | Exhaustive rational coboundary obstruction                      | Exact for real $\omega > 0$ |
| Cocycle and local critical-mass normal form    | Exact substitution in $\mathbb{C}(r, \omega)$                   | Exact for $\omega \neq 0$   |
| Endpoint normalization of the mass comparison  | Explicit horizon/infinity Wronskian transgression               | Open                        |
| Defective physical specialization              | Winding-one contour, local selector, spin-one unit              | Validated theorem           |
| Non-Einstein generalized root                  | Defective Smith type and exact filtration                       | Exact consequence           |
| Exterior cut-off outgoing Green double pole    | Analytic Jost maps, transparent cuts, finite-interval reduction | Exact/validated theorem     |
| Causal spacetime pole and generalized ringdown | Global weighted Fredholm and contour theory                     | Not claimed                 |
| Physical Einstein–Weyl mass identification     | Finite-mass cocycle and endpoint normalization                  | Open                        |

## 2 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \quad D = f \partial_r, \quad f = 1 - \frac{2}{r},$$

with  $\omega \neq 0$  unless a threshold limit is stated explicitly. All rational gauges and module morphisms are over the differential field  $(K, D)$ . We use  $D(q)$ ,  $D(U)$ , and  $D^3q$ , rather than primes, to avoid confusing the tortoise derivative with  $\partial_r$ .

The axial spin-two and spin-one scalar operators are

$$L_2 = D^2 + U_2, \quad U_2 = \omega^2 - f \left( \frac{6}{r^2} - \frac{6}{r^3} \right),$$

$$L_1 = D^2 + U_1, \quad U_1 = \omega^2 - f \frac{6}{r^2}.$$

The latter is the  $\ell = 2$  Maxwell Regge–Wheeler equation. A rank-two scalar module is represented by its companion system on  $(y, Dy)^T$ .

The source Einstein layer is the kernel of the linearized Ricci/Schouten carrier map. The target spin-two quotient is its physical Regge–Wheeler carrier. The third quotient is the Maxwell class of the target Einstein gauge vector. Consequently, “non-Einstein” below means nonzero projection to the gauge-invariant carrier quotient, not failure of a coordinate gauge condition.

## 3 Exact axial RW/RW/Maxwell filtration

In the certified factor gauge the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1 Z, \end{aligned} \tag{1}$$

where  $X, Y, Z \in K^2$ , the companion matrices  $A$  and  $A_1$  represent  $L_2$  and  $L_1$ , and the forcing blocks factor through a single scalar source:

$$E = USJ, \quad C = USN.$$

The rational reconstruction has full row rank on the declared domain.

**Theorem 3.1** (Filtered axial Bach module). *For axial  $\ell = 2$  Schwarzschild perturbations, the reduced six-state Bach module has a filtration whose successive simple factors are*

$$M_2, \quad M_2, \quad M_1.$$

*The middle two-spin-two block is an extension  $E_2$  of  $M_2$  by  $M_2$ . The quotient  $M_1$  is the Maxwell module of the carrier gauge vector. The filtration is exact but is not asserted to be a direct sum.*

The theorem is an exact rational statement. It is independent of endpoint normalization, scattering conventions, and the later QNM specialization.

## 4 Partial-jet realization

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}.$$

Let its fundamental matrix be

$$\Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

**Theorem 4.1** (Exact partial jet). *The six-state generator in (1) is the first spin-two-row jet of  $\mathcal{B}(\tau)$  at  $\tau = 0$ . Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \quad \det \Phi_6 = (\det P)^2 \det R.$$

Consequently any factor-adapted endpoint connection has the triangular form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \quad \det T = a^2 g.$$

*Proof.* Differentiate  $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$  at  $\tau = 0$ . The equations for  $P, \dot{P}, Q, \dot{Q}, R$  reproduce the three rows of (1) column by column. The determinant identity follows from block triangularity. The same identity applied to endpoint transition matrices gives the displayed connection form.  $\square$

The parameter  $\tau$  is intrinsic at this stage. Identifying it with a physical Fierz–Pauli mass requires an additional finite-mass reduction and compatible endpoint germs; Section 15 records the exact remaining gate.

## 5 Projective gauge and symmetric-square cohomology

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \quad Ly = 0, \quad L = D^2 + U. \quad (2)$$

Set

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \quad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

**Lemma 5.1** (Explicit triangular gauge). *For  $q \in K$ , define*

$$Q(q) = qD - \frac{1}{2}D(q).$$

*On every solution  $Ly = 0$ ,*

$$[L, Q(q)]y = -\frac{1}{2}\mathcal{K}_U(q)y.$$

*Thus the change of lift  $x \mapsto x + Q(q)y$  changes the scalar extension density by*

$$\mathcal{I} \mapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

*Proof.* Expand  $LQ(q) - Q(q)L$ , use the Leibniz rule for  $D$ , and then set  $D^2y = -Uy$ . The coefficient of  $Dy$  cancels, while the coefficient of  $y$  is  $-\frac{1}{2}[D^3q + 4UDq + 2D(U)q]$ .  $\square$

The extension is therefore represented by

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

Because Lemma 5.1 contains the factor  $1/2$ , removal of a term  $\mathcal{K}_U q$  uses the triangular gauge  $Q(2q)$ . This factor does not alter the cohomology quotient but matters in an explicit frame calculation.

The equation  $\mathcal{K}_U q = 0$  is the symmetric-square equation of  $L$ . If  $y_1, y_2$  span the scalar solution space, then its solution space is spanned by  $y_1^2, y_1 y_2, y_2^2$ . The same operator controls both the projective obstruction and trace-free horizontal endomorphisms.

## 6 Mass-direction normal form of the Bach cocycle

For the axial  $\ell = 2$  Bach extension, rational reduction gives

$$\mathcal{I}_{\text{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

**Theorem 6.1** (Mass-direction normal form). *Let*

$$q(r, \omega) = -\frac{i}{120\omega} \left( 15r + 13 + \frac{12}{r} + \frac{9}{r^2} \right).$$

For  $\omega \neq 0$ ,

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2} q + \frac{i\omega}{2} f.$$

Consequently

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} [f] \quad \text{in} \quad K/\mathcal{K}_{U_2} K.$$

In particular,  $[f] \neq 0$  wherever the Bach self-extension is non-split.

*Proof.* Apply  $D = f\partial_r$  three times to  $q$  and substitute

$$U_2 = \omega^2 - f \left( \frac{6}{r^2} - \frac{6}{r^3} \right)$$

in  $\mathcal{K}_{U_2} q = D^3q + 4U_2 Dq + 2D(U_2)q$ . Clearing the common denominator  $120r^4\omega$  reduces the claim to a polynomial identity in  $r$  and  $\omega$ . Exact coefficient comparison gives the displayed formula. The quotient identity follows immediately. If  $[f] = 0$ , then  $[\mathcal{I}_{\text{Bach}}] = 0$ , which would split the extension.  $\square$

*Remark 6.2* (Threshold scope). The displayed  $1/\omega$  singularity of  $\mathcal{I}_{\text{Bach}}$  is cohomologically removable: its intrinsic class has the representative  $(i\omega/2)f$ . The removing gauge  $Q(2q)$ , however, is itself  $O(\omega^{-1})$ . The theorem therefore does not construct a threshold-uniform splitting or justify commuting the limits  $\omega \rightarrow 0$  and  $\tau \rightarrow 0$ .

*Remark 6.3* (Mass-shaped is not yet physical mass). The representative  $f = 1 - 2/r$  is the radial direction expected from a mass after a helicity-two reduction. The theorem proves a local cohomology normal form, not equality with a particular Einstein–Weyl mass parameter. If a finite-mass reduction gives

$$\mathcal{I}_{\text{mass}} = c(\omega)f + \mathcal{K}_{U_2} p,$$

then the two extensions are locally equivalent precisely when  $c(\omega) \neq 0$ ; their parameter normalization is  $2c(\omega)/(i\omega)$ .

## 7 Period formula for the connection shear

Let  $y_1, y_2$  be a scalar basis and set

$$Y = \begin{pmatrix} y_1 & y_2 \\ Dy_1 & Dy_2 \end{pmatrix}, \quad W = y_1 Dy_2 - y_2 Dy_1.$$

Because the companion matrix has zero trace,  $D(W) = 0$ .

**Theorem 7.1** (Symmetric-square period matrix). *For the family*

$$(D^2 + U + \tau V)y = 0,$$

*the interaction-picture variation satisfies*

$$Y^{-1} \partial_\tau A Y = \frac{V}{W} \begin{pmatrix} y_1 y_2 & y_2^2 \\ -y_1^2 & -y_1 y_2 \end{pmatrix}.$$

*Consequently the spin-two connection shear is determined by the three periods*

$$\int \frac{V}{W} y_1^2 dr_*, \quad \int \frac{V}{W} y_1 y_2 dr_*, \quad \int \frac{V}{W} y_2^2 dr_*.$$

*Proof.* In the basis  $(y, Dy)^T$ ,

$$A(\tau) = \begin{pmatrix} 0 & 1 \\ -(U + \tau V) & 0 \end{pmatrix}, \quad \partial_\tau A = \begin{pmatrix} 0 & 0 \\ -V & 0 \end{pmatrix}.$$

Insert

$$Y^{-1} = \frac{1}{W} \begin{pmatrix} Dy_2 & -y_2 \\ -Dy_1 & y_1 \end{pmatrix}$$

and multiply. Integration of the interaction-picture equation gives the three entries.  $\square$

A coboundary  $\mathcal{K}_U q$  changes these periods by endpoint terms. This is the connection-level form of Lemma 5.1: rational triangular gauges alter Jost frames but not the intrinsic divisor or Smith selector once the endpoint units are tracked.

## 8 Nonsplitting, endomorphisms, and Fitting ideals

**Theorem 8.1** (Non-split physical-axis extension). *For every real  $\omega > 0$ , the axial  $\ell = 2$  spin-two block*

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0$$

*is non-split over  $\mathbb{C}(r)[D]$ .*

*Certificate-supported exact proof.* The rational splitting equation reduces, after exhaustive pole and degree bounds, to a finite coefficient system. Fixed minors of its coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456 \omega^{10}, \quad \det[M \mid \text{rhs}]_{[0,1,2,5]} = -645120 i \omega^9.$$

They are simultaneously nonzero for every real  $\omega > 0$ , so the inhomogeneous coboundary equation is inconsistent. The fixed-minor argument also covers the former frame event  $\omega^2 = 3$ .  $\square$

The scalar Regge–Wheeler module is simple on the positive real axis and has scalar rational endomorphism ring. The only nonzero complex reducibility points allowed by the terminal Riccati ansatz with matching horizon and infinity signs are the algebraically special frequencies

$$\omega = \pm i \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2)}{12}.$$

For  $\ell = 2$ , these are  $\omega = \pm 2i$ , away from the physical axis.

**Theorem 8.2** (Local commutant). *For real  $\omega > 0$ ,*

$$\text{End}_{\mathbb{C}(r)[D]}(E_2) \simeq \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

*Every endomorphism is  $aI + bN$ , where  $N^2 = 0$ . Hence the only idempotents are  $0, I$ , the only involutions are  $\pm I$ , and every diagonalizable local endomorphism is scalar.*

*Proof.* The submodule  $M_2$  is the unique simple submodule: a second simple submodule mapping nontrivially to the quotient would split the extension. An endomorphism therefore induces scalars  $a, d$  on the submodule and quotient. Naturality of the nonzero extension class gives  $a = d$ . Subtracting  $aI$  leaves a map that vanishes on the submodule and quotient, so it factors through the quotient back into the submodule and is a scalar multiple of the canonical nilpotent  $N$ . The corollaries follow from  $(aI + bN)^k = a^k I + k a^{k-1} b N$ .  $\square$

For the triangular connection

$$T = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix},$$

the Fitting ideals are

$$I_1 = (a, b, c, d, g), \quad I_2 = (a^2, ad, bd - ac, ag, bg), \quad I_3 = (a^2 g).$$

They give a normalization-independent local classification of all specializations.

## 9 Analytic Jost connections and Smith dichotomy

The moving Frobenius/Jost phases define analytic endpoint-selected planes on the certified spectral domains. Transport to a common ordinary radius and change to a factor-adapted frame. At a simple spin-two QNM  $\omega_n$ , assume the spin-one coefficient  $g(\omega_n) \neq 0$ . Analytic row and column operations remove the spin-one row and column, leaving

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \quad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

**Proposition 9.1** (Local Smith dichotomy). *At  $\omega_n$ ,*

$$\text{Smith}(T) = \begin{cases} (0, 0, 2), & b(\omega_n) \neq 0, \\ (0, 1, 1), & b(\omega_n) = 0. \end{cases}$$

*The first case has geometric multiplicity one and one length-two root chain; the second has two independent length-one roots.*

*Proof.* Let  $z = \omega - \omega_n$ . Since  $a = a_1 z + O(z^2)$  with  $a_1 \neq 0$ , the determinant valuation of the two-spin-two block is two. If  $b(0) \neq 0$ , its first Fitting ideal is the unit ideal, so its exponents are  $(0, 2)$ . If  $b(0) = 0$ , all entries vanish to first order and the exponents are  $(1, 1)$ . Appending the spin-one unit adds a zero exponent.  $\square$

Projectively, write the selected endpoint lines in a common chart as

$$v_H(\omega, \tau), \quad v_\infty(\omega, \tau), \quad \delta = v_\infty - v_H.$$

At a simple root,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n, 0)}{\delta_\omega(\omega_n, 0)}.$$

This ratio is invariant under analytic nonzero endpoint units and under moving the matching section.

## 10 Certified QNM specialization

The validated contour is oriented positively and encloses the disk

$$\Omega_n = \{\omega : |\omega - (-0.3736716844180418 + 0.0889623156889357 i)| \leq 10^{-7}\}.$$

Panelwise projective enclosures exclude zero on the contour, include all chart-transition units, and give winding number +1. Thus the disk contains exactly one simple spin-two zero.

On the localized disk, the intrinsic tangent is enclosed by

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation gives a spin-one mismatch modulus greater than  $1/2500$ .

**Theorem 10.1** (Certified defective Schwarzschild resonance). *The complete axial connection has Smith valuations*

$$(0, 0, 2)$$

*at the unique QNM in  $\Omega_n$ . The algebraic multiplicity of its spin-two divisor is two, its geometric multiplicity is one, and it has one length-two connection root chain.*

*Proof.* The winding certificate gives one simple spin-two zero. The local tangent enclosure proves  $b(\omega_n) \neq 0$ , while the spin-one enclosure proves  $g(\omega_n) \neq 0$ . Proposition 9.1 applies.  $\square$

## 11 Resonant evaluation of the self-extension

The projective selector, adjoint overlap, root-chain coefficient, and Green-pole residue are different representations of one resonant functional. This is the finite-interval Evans/Melnikov mechanism; compare the general relation between Evans derivatives and adjoint generalized eigenfunctions in [1].

Let

$$L(\omega) : X \longrightarrow Y$$

denote the analytic finite-interval spin-two boundary pencil, with the endpoint conditions included either in its domain or as finite-dimensional components of an augmented range. At a simple resonance choose right and left states

$$L_n u_n = 0, \quad L_n^* \tilde{u}_n = 0, \quad L_n = L(\omega_n),$$



and define, using the bilinear duality of the augmented boundary problem,

$$\alpha_n = \langle \tilde{u}_n, L'(\omega_n)u_n \rangle, \quad \alpha_n \neq 0.$$

Let the repeated block be

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix}, \quad \beta_n = \langle \tilde{u}_n, K(\omega_n)u_n \rangle.$$

All endpoint components of the augmented pencil and its duality are included in these pairings.

**Theorem 11.1** (Resonant evaluation theorem). *In endpoint frames compatible with the partial jet,*

$$\boxed{\kappa_n := \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = -\omega'_n(0).}$$

Here  $\omega_n(\tau)$  is the resonance branch of

$$L(\omega, \tau) = L(\omega) + \tau K(\omega)$$

through  $\omega_n$ . The equality is invariant under a common analytic nonzero Evans unit.

*Proof.* Differentiate

$$L(\omega_n(\tau), \tau)u(\tau) = 0$$

at  $\tau = 0$ :

$$L_n \dot{u} + \omega'_n(0)L'_n u_n + K_n u_n = 0.$$

Pairing with  $\tilde{u}_n$  removes the first term and gives

$$\omega'_n(0)\alpha_n + \beta_n = 0.$$

On the connection side, differentiation of

$$a(\omega_n(\tau), \tau) = 0$$

gives

$$a'(\omega_n)\omega'_n(0) + \partial_\tau a(\omega_n, 0) = 0.$$

The exact partial-jet construction identifies  $\partial_\tau a(\omega_n, 0) = b(\omega_n)$ . Multiplication of the Evans coefficient by an analytic unit  $u(\omega, \tau)$  multiplies both  $a'$  and  $b$  at the root by  $u(\omega_n, 0)$ , so the ratio is unchanged.  $\square$

**Proposition 11.2** (Resonant functional on the extension class). *The assignment*

$$\mathfrak{M}_n([K]) := \langle \tilde{u}_n, K(\omega_n)u_n \rangle$$

*is a well-defined linear functional on the local self-extension class. Thus nonsplitting is the statement  $[K] \neq 0$ , while defectiveness at  $\omega_n$  is the sharper statement  $\mathfrak{M}_n([K]) \neq 0$ .*

*Proof.* Under a triangular change of splitting,

$$K \mapsto K + LQ - QL.$$

At the resonance,

$$\langle \tilde{u}_n, (L_n Q - Q L_n)u_n \rangle = \langle L_n^* \tilde{u}_n, Q u_n \rangle - \langle \tilde{u}_n, Q L_n u_n \rangle = 0.$$

When endpoint conditions are encoded in an augmented pencil, the same calculation includes the corresponding boundary terms by definition.  $\square$

The theorem and proposition separate a global module property from its evaluation at one resonance. A nonzero extension class need not be defective at every QNM: the linear functional  $\mathfrak{M}_n$  may vanish at isolated modes.

## 12 The generalized root vector is non-Einstein

**Proposition 12.1** (Nonzero carrier in the generalized root). *Assume*

$$a(\omega_n) = 0, \quad a'(\omega_n) \neq 0, \quad b(\omega_n) \neq 0.$$

*The geometric root vector belongs to the source Einstein submodule, while every length-two root polynomial has first generalized coefficient with nonzero projection to the carrier quotient. In the normalized triangular frame that projection is*

$$-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n}.$$

*Thus the defective doubled resonance is not two copies of an Einstein quasinormal mode: its generalized member is intrinsically non-Einstein.*

*Proof.* Put  $z = \omega - \omega_n$  and write

$$a = a_1 z + O(z^2), \quad b = b_0 + O(z), \quad a_1 b_0 \neq 0.$$

For

$$c(z) = c_0 + z c_1 + O(z^2), \quad T_2(z) c(z) = O(z^2),$$

the order-zero equation gives  $c_0 = e_X$ , the source Einstein vector. Write  $c_1 = x e_X + y e_Y$ . The  $e_X$  component at order  $z$  is

$$a_1 + b_0 y = 0,$$

so  $y = -a_1/b_0 \neq 0$ . By Theorem 3.1,  $e_Y$  projects nontrivially to the Ricci-carrier quotient. Adding a multiple of  $e_X$  changes  $x$ , but cannot remove  $y$ .  $\square$

## 13 Exterior cut-off outgoing Green double pole

Fix ordinary radii  $2 < r_H < r_I < \infty$  inside the certified analytic continuation domains. Let

$$H(\omega) : \mathbb{C}^3 \longrightarrow C^\infty([r_H, r_I]; \mathbb{C}^6)$$

be the analytic Poisson operator for horizon-selected homogeneous data, let  $\Gamma_-(\omega)$  be the unwanted incoming trace at  $r_I$ , and set

$$T(\omega) = \Gamma_-(\omega) H(\omega).$$

For a compactly supported source  $F$ , let  $P_H(\omega)F$  denote the inhomogeneous solution with zero horizon data. The outgoing solution is

$$R_{\text{out}}(\omega)F = P_H(\omega)F - H(\omega)T(\omega)^{-1}\Gamma_-(\omega)P_H(\omega)F. \quad (3)$$

**Theorem 13.1** (Finite-interval outgoing Green pole). *On a neighborhood of the certified frequency  $\omega_n$ , the operator*

$$R_{\text{out}}(\omega) : C_c^\infty((r_H, r_I); \mathbb{C}^6) \longrightarrow C_{\text{loc}}^\infty((r_H, r_I); \mathbb{C}^6)$$

*is meromorphic and has a nonzero rank-one pole of order two. Up to the endpoint trace sign convention, its principal coefficient is*

$$\text{Coeff}_{(\omega - \omega_n)^{-2}} R_{\text{out}} = \frac{b(\omega_n)}{a'(\omega_n)^2} H(\omega_n) e_X \otimes [e_Y^* \Gamma_-(\omega_n) P_H(\omega_n)].$$

*Proof.* All factors in (3) except  $T^{-1}$  are analytic. After analytic elimination of the spin-one unit, the singular block is

$$T_2^{-1} = \begin{pmatrix} a^{-1} & -ba^{-2} \\ 0 & a^{-1} \end{pmatrix},$$

and therefore

$$\text{Coeff}_{z^{-2}} T_2^{-1} = -\frac{b(\omega_n)}{a'(\omega_n)^2} e_X \otimes e_Y^*.$$

The minus sign in (3) gives the displayed convention.

The range vector is nonzero because the homogeneous Poisson map is injective, so  $H(\omega_n)e_X \neq 0$ . The covector is nonzero by a cutoff Jost construction. Given any incoming coefficient  $q$ , choose the corresponding incoming Jost solution  $J_-q$  near  $r_I$ , multiply it by a smooth cutoff  $\chi$  that vanishes near  $r_H$  and equals one near  $r_I$ , and set

$$F = (D - \mathbb{A}(\omega_n))(\chi J_-q).$$

Then  $F$  is compactly supported,  $P_H(\omega_n)F = \chi J_-q$ , and  $\Gamma_- P_H(\omega_n)F = q$ . Hence the incoming trace map from compact sources is onto. Taking  $q = e_Y$  shows that the covector does not vanish. The tensor product is therefore nonzero and rank one.  $\square$

### 13.1 Exact transparent cuts

Put  $x = r_*$ . On the certified QNM neighborhood, let

$$u_H(x, \omega), \quad u_\infty(x, \omega)$$

be analytic spin-two Jost solutions satisfying the future-horizon and outgoing-infinity conditions. The scalar Evans coefficient is, up to an analytic unit,

$$a(\omega) = W(u_H, u_\infty).$$

Let  $\chi_s$  and  $\chi_o$  be nontrivial smooth source and observation cutoffs with compact support. Choose finite cuts

$$x_- < \text{supp } \chi_s \cup \text{supp } \chi_o < x_+.$$

At the two cuts impose the exact transparent boundary conditions

$$W(u, u_H)|_{x_-} = 0, \quad W(u, u_\infty)|_{x_+} = 0.$$

Their coefficients are analytic in  $\omega$ . Uniqueness of the exterior Jost problems shows that the resulting finite-interval inverse agrees exactly, between the cutoffs, with the meromorphically continued exterior outgoing inverse.

**Theorem 13.2** (Exterior cut-off Green pole). *For any nontrivial compactly supported source and observation cutoffs,*

$$\chi_o R_{\text{ext}}(\omega) \chi_s$$

*has a genuine nonzero rank-one pole of order two at the certified Schwarzschild QNM. In the augmented Fredholm normalization its principal coefficient is*

$$\text{Coeff}_{(\omega - \omega_n)^{-2}} (\chi_o R_{\text{ext}} \chi_s) = -\frac{\beta_n}{\alpha_n^2} (\chi_o u_n) \otimes (\tilde{u}_n \chi_s).$$

*Proof.* The transparent conditions select exactly the restrictions of the exterior horizon and infinity Jost solutions. Hence

$$\chi_o R_{\text{ext}}(\omega) \chi_s = \chi_o R_{[x_-, x_+]}(\omega) \chi_s.$$

Theorem 13.1 and (4) give the displayed principal coefficient. It is nonzero:  $\beta_n \neq 0$ , while a nonzero solution of a second-order ODE cannot vanish on a nonempty open interval. Thus  $\chi_o u_n$  and  $\tilde{u}_n \chi_s$  are nonzero for every nontrivial smooth cutoff.  $\square$

**Corollary 13.3** (Non-annihilation by certified reconstruction). *The certified rational reconstruction into the axial metric/curvature variables does not annihilate the range  $H(\omega_n)e_X$ . Hence at least one reconstructed local observable has the same order-two radial response pole.*

In the finite-interval Fredholm normalization of Section 11, the scalar resolvent has

$$L(\omega)^{-1} = \frac{u_n \otimes \tilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block of  $\mathbb{L}^{-1}$  is  $-L^{-1}KL^{-1}$ , and Theorem 11.1 therefore gives the invariant principal coefficient

$$\boxed{\text{Coeff}_{(\omega - \omega_n)^{-2}}(\mathbb{L}^{-1})_{12} = -\frac{\beta_n}{\alpha_n^2} u_n \otimes \tilde{u}_n = -\frac{\kappa_n}{\alpha_n} u_n \otimes \tilde{u}_n.} \quad (4)$$

The Poisson/trace coefficient in Theorem 13.1 and the exterior coefficient in Theorem 13.2 are the same rank-one residue expressed in endpoint and augmented Fredholm coordinates.

## 14 Conditional global Fredholm promotion

Theorem 13.2 is an exterior differential Green-operator theorem after compact source and observation localization. A global physical resolvent without cutoffs requires a separate closed-operator realization. The correct abstract block notation is

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

where  $L(\omega) : D \subset X \rightarrow Y$  is an analytic Fredholm pencil of index zero and  $K(\omega) : D \rightarrow Y$  is graph-norm bounded. Suppose

$$u_n \in \ker L(\omega_n), \quad \tilde{u}_n \in \ker L(\omega_n)^*,$$

and define using the  $Y^*$ - $Y$  bilinear duality

$$\alpha_n = \langle \tilde{u}_n, L'(\omega_n)u_n \rangle, \quad \beta_n = \langle \tilde{u}_n, K(\omega_n)u_n \rangle.$$

**Proposition 14.1** (Conditional global pole formula). *If  $L^{-1}$  is meromorphic near  $\omega_n$ , its zero is simple,  $\alpha_n \neq 0$ , and  $\beta_n \neq 0$ , then*

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1}KL^{-1} \\ 0 & L^{-1} \end{pmatrix}$$

*has an order-two pole with principal coefficient*

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \tilde{u}_n$$

*in the upper-right block.*

This proposition is standard block algebra once the Fredholm spaces and pairing exist. What remains open is not the exterior cut-off pole proved above, but the construction of:

1. closed weighted horizon and null-infinity domains;
2. a physical gauge quotient and bounded metric reconstruction;
3. a causal exterior resolvent on a Laplace contour;
4. a justified contour deformation and control of the non-pole part.

Only after these steps could the double pole be promoted to a rigorous  $te^{i\omega_n t}$  contribution to retarded ringdown.

## 15 Critical Einstein–Weyl mass comparison

On the transverse-traceless spin-two sector, the quadratic Einstein–Weyl parent equation is

$$(\mathcal{E} + m\mathcal{A})\phi = 0,$$

where the parameter  $m$  has the dimensions of squared mass. Its Schwarzschild axial scalar reduction is

$$L_m(\omega)y = [D^2 + \omega^2 - V_2 - mf]y = 0, \quad f = 1 - \frac{2}{r}. \quad (5)$$

For a differentiable family  $y(m)$ , differentiation at  $m = 0$  gives

$$L_2 \partial_m y = f y.$$

**Theorem 15.1** (Exact local critical-mass-jet identification). *The projective cocycle of the transverse-traceless critical mass jet is*

$$[\mathcal{I}_{\text{mass}}] = [f].$$

Consequently

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} [\mathcal{I}_{\text{mass}}].$$

With the convention  $L_m = L_2 - mf$ , the local first-order parameter relation is

$$m = \frac{i\omega}{2} \tau$$

modulo rational triangular gauge.

*Proof.* Equation (5) gives the extension equation  $L_2 \partial_m y = f y$ , so its scalar projective density is  $f$ . Theorem 6.1 gives  $[\mathcal{I}_{\text{Bach}}] = (i\omega/2)[f]$ . Comparing the two tangent equations yields the parameter relation.  $\square$

This theorem closes the local transverse-traceless differential-module comparison. It does not identify independently normalized global Jost connections, because the equivalence gauge is not bounded at infinity.

## 15.1 Forced asymptotics of the equivalence gauge

The rational representative in Theorem 6.1 satisfies

$$q(r, \omega) = -\frac{i}{8\omega}r + O(1), \quad r \rightarrow \infty.$$

**Proposition 15.2** (No bounded rational mass-equivalence gauge). *Every rational  $q$  satisfying*

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2}q + \frac{i\omega}{2}f$$

*has the forced leading behavior*

$$q(r, \omega) = -\frac{i}{8\omega}r + O(1).$$

*In particular, no bounded rational triangular gauge identifies the Bach jet with the critical mass jet at infinity.*

*Proof.* At infinity,

$$\mathcal{I}_{\text{Bach}} - \frac{i\omega}{2}f \longrightarrow -\frac{i\omega}{2}, \quad U_2 \longrightarrow \omega^2.$$

A rational  $q = o(r)$  is  $O(1)$  and gives  $\mathcal{K}_{U_2}q = o(1)$ , so it cannot produce the nonzero constant. If  $q \sim cr$ , then  $Dq \rightarrow c$ , while  $D^3q \rightarrow 0$  and  $D(U_2)q \rightarrow 0$ . Therefore

$$\mathcal{K}_{U_2}q \longrightarrow 4\omega^2c.$$

Matching the constant gives  $c = -i/(8\omega)$ , uniquely.  $\square$

## 15.2 Boundary transgression

The normalization in Lemma 5.1 must be retained here. Put

$$Q(q) = qD - \frac{1}{2}D(q), \quad \widehat{Q} = -2Q(q) = -2qD + D(q).$$

Because  $[L_2, Q(q)]|_{\ker L_2} = -\frac{1}{2}\mathcal{K}_{U_2}(q)$ , the bulk extension operators obey

$$K_{\text{Bach}} - \frac{i\omega}{2}K_{\text{mass}} = [L_2, \widehat{Q}] \tag{6}$$

in the scalarized companion convention. The factor  $-2$  is essential: using  $Q(q)$  itself would be inconsistent with the cocycle normalization of Theorem 6.1.

**Theorem 15.3** (Finite-cut boundary transgression). *Let  $u_n, \tilde{u}_n$  be resonant right and left states, with the finite-cut pairing defined by Green's formula. Then*

$$\boxed{\beta_{n,[x_-,x_+]}^{\text{Bach}} - \frac{i\omega_n}{2}\beta_{n,[x_-,x_+]}^{\text{mass}} = \left[W(\tilde{u}_n, \widehat{Q}u_n)\right]_{x_-}^{x_+}.$$

*In an augmented boundary pencil, the same term is absorbed into the derivatives of the horizon and infinity boundary maps, restoring splitting-gauge invariance of the complete overlap.*

*Proof.* On finite cuts, use (6),  $L_2 u_n = 0$ , and  $L_2^\dagger \tilde{u}_n = 0$ :

$$\begin{aligned} \left\langle \tilde{u}_n, \left( K_{\text{Bach}} - \frac{i\omega_n}{2} K_{\text{mass}} \right) u_n \right\rangle &= \left\langle \tilde{u}_n, [L_2, \hat{Q}] u_n \right\rangle \\ &= \left[ W(\tilde{u}_n, \hat{Q} u_n) \right]_{x_-}^{x_+}. \end{aligned}$$

This proves the finite-cut identity. Equivalently, include the boundary functionals in the augmented operator and its duality before differentiation.  $\square$

Theorem 15.1 and Proposition 11.2 give

$$\beta_n^{\text{Bach}} = \frac{i\omega_n}{2} \mathfrak{M}_n([f]), \quad \mathfrak{M}_n([f]) \neq 0.$$

After the boundary transgression has been evaluated in matched physical Jost frames, the predicted mass slope is

$$\left. \frac{d\omega_n}{dm} \right|_{m=0} = \frac{2i}{\omega_n} \kappa_n^{\text{Bach}}.$$

This final equality is not yet claimed: the local bulk comparison is exact, but the remaining physical normalization is exactly the endpoint limit

$$\left[ W(\tilde{u}_n, \hat{Q} u_n) \right]_{\mathcal{H}^+}^{\mathcal{I}^+},$$

which remains to be computed or regularized in matched Jost frames.

### Exact parent/radial comparison target

Let  $R$  reconstruct the tensor carrier from the scalar master field and let  $P$  project the tensor equation back to the scalar equation. The corresponding exact target is

$$K_{\text{radial}} - PAR = LQ_0 - Q_0L$$

in the augmented pencil, with its endpoint component recorded explicitly. Once proved, Proposition 11.2 identifies the parent and radial overlaps without a separate numerical equality test.

## 16 Computational certificates and claim boundary

The computer-assisted inputs are separated from the exact derivations above.

| Certificate family         | Claim used here                                                      | Evidence type                         |
|----------------------------|----------------------------------------------------------------------|---------------------------------------|
| Axial triangular preflight | Exact RW/RW/Maxwell filtration and reconstruction                    | Exact rational                        |
| Projective cocycle         | Gauge law, nontrivial extension class, mass normal form              | Exact rational                        |
| Critical parent mass jet   | TT mass variation and difference quotient                            | Exact covariant/reduced               |
| Analytic continuation      | Analytic Jost germs on the declared QNM domain                       | Exact continuation gate               |
| RW/Maxwell simplicity      | Physical-axis simplicity and scalar endomorphisms                    | Exact rational                        |
| Full Evans contour         | Zero-free boundary and winding +1                                    | Validated interval                    |
| Local selector             | $\delta_\tau \neq 0$ on the root disk                                | Validated interval                    |
| Spin-one local unit        | Full three-factor Smith type $(0, 0, 2)$                             | Validated interval                    |
| Fredholm promotion         | Finite-interval rank-one radial Green pole and transparent-cut input | Exact reduction plus validated inputs |

The theorem statements and proofs are the mathematical authority. The machine-readable records pin exact witnesses, interval margins, source hashes, and independently executable checks. A missing input, unsupported domain, timeout, or failed mutation causes rejection rather than partial acceptance.

**Explicit non-results.** This paper does not establish a causal spacetime resolvent, a complete QNM expansion, a late-time asymptotic formula, a  $te^{i\omega_n t}$  term, a time-domain instability, equality of the globally normalized Bach and massive Jost slopes, an evaluated endpoint transgression, an all- $\ell$  Bach nonsplitting theorem, or a quantum positivity statement.

## 17 Conclusion

The axial Schwarzschild Bach equation contains a repeated Regge–Wheeler factor, but the repetition is not a rational direct sum. It is a non-split self-extension whose projective class is represented by the elementary direction  $(i\omega/2)(1 - 2/r)$ . That direction is exactly the transverse-traceless critical mass jet. The equivalence gauge has forced linear growth at infinity, so the remaining global normalization question is one explicit endpoint Wronskian. At the certified damped mode, the scalar zero is simple while the extension selector is nonzero. The resulting defect has concrete differential content: the generalized root has a nonzero Ricci-carrier quotient, and the compactly cut-off exterior outgoing Green operator has a genuine rank-one double pole.

This separates three notions that should not be conflated. A repeated scalar divisor fixes algebraic multiplicity; the extension selector decides whether that multiplicity is semisimple or defective; and a causal time-domain interpretation requires a further global Fredholm theory. The first two questions are settled here at one physical Schwarzschild resonance. The third remains a sharply delimited next problem.



## References

- [1] T. Kapitula, The Evans function and generalized Melnikov integrals, *SIAM Journal on Mathematical Analysis* **30** (1999), 273–297, [doi:10.1137/S0036141097327963](https://doi.org/10.1137/S0036141097327963).